Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

Applied AI engineer who **ships production Claude-powered systems** - the exact work Anthropic's FDE team delivers. Built AgentProbe end-to-end: **production LLM agents** (3-agent LangGraph system), **multi-provider routing** (Anthropic, OpenAI, Groq, Ollama), and a **5-metric evaluation framework** with CI/CD regression gates - the complete stack an FDE deploys at strategic accounts. Built AI on FHIR: HIPAA-compliant clinical NLP platform (BERT, spaCy, FHIR APIs) - direct proof in **healthcare**, one of Anthropic's preferred enterprise verticals. **8+ years production Python**; enterprise-scale delivery at Entrata (100K+ MAU, 120+ Agile sprints, 1.6M financial records with OpenAI API). M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026). *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

Production LLM Agents (LangGraph, ReAct) | Prompt & Context Engineering | LLM Evaluation Frameworks (LLM-as-Judge) | Multi-Provider LLM Routing (Anthropic, OpenAI, Groq) | RAG Pipelines & Vector Search (ChromaDB) | Python (8+ years production) | Enterprise-Scale Delivery (100K+ MAU, 120+ sprints) | Healthcare AI / HIPAA Compliance | FastAPI / REST APIs / Full-Stack | AWS / Docker / CI/CD (GitHub Actions) | Cross-Functional Technical Collaboration | Open-Source Development

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Applied **Python, PyTorch, and OpenAI API** to automate data transformation across **1.6M financial records** - production AI on regulated enterprise data with output validation and quality monitoring.
- Built and delivered **20+ full-stack modules** (Python, TypeScript/JavaScript, PostgreSQL) across **120+ Agile sprints** for **100K+ monthly active users** - shipped production systems end-to-end from scoping through deployment for enterprise customers.
- Collaborated cross-functionally with product managers, business counterparts, and cross-squad teams to translate requirements into production systems - the same technical-to-business translation FDEs perform at customer sites.
- Led Docker-based containerization for 30%+ of microservices, cutting execution time by **50%**; improved production query performance by **30%**; maintained **99.5% uptime** for 6+ months across enterprise SaaS.
- Migrated client data to **AWS S3** with Libsodium-encrypted pipelines - **100% PHI compliance, zero security incidents**; built 5+ internal platform tools adopted across squads, reducing resolution time by **40%**.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Mentored 30+ students; translated complex technical concepts into accessible explanations - the same communication and discovery pattern FDEs apply when working with enterprise stakeholders.

PROJECTS

AgentProbe: Production Claude-Compatible Agent Platform - LLMs, RAG, Evaluation OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Anthropic · OpenAI · Groq · Ollama · Docker · GitHub Actions CI/CD

Built the complete stack an Anthropic FDE deploys at strategic accounts - production agents, evaluation frameworks, and LLM routing - from greenfield to open-source. **Production LLM agents**: 3-agent LangGraph system (Orchestrator, RAG Tracer, Auto Fix) using ReAct patterns - multi-step reasoning, tool calls, failure handling, inter-agent state handoffs. **Multi-provider routing**: Anthropic, OpenAI, Groq, Ollama - dynamic selection by task type, latency, and capability. Primary provider is Anthropic (Claude). **Evaluation framework**: 5-metric LLM-as-judge engine (factual accuracy, hallucination, retrieval, completeness, compliance) with per-run regression tracking and CI/CD gates - agents are reliable, observable, and auditable. **RAG pipeline**: ChromaDB + embeddings + retrieval quality scoring across multi-step agent runs. **Platform layer**: 15-endpoint FastAPI backend + React observability dashboard - the shared abstraction FDEs document as repeatable deployment patterns. *github.com/venkatalakshmiparimala/AgentProbe*

AI on FHIR: Healthcare AI Platform - LLMs + FHIR APIs (HIPAA-Compliant)

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings

Production AI for the healthcare vertical - the same enterprise domain Anthropic's FDE team prioritizes. NLP platform routing natural language queries to FHIR APIs via LLMs, BERT, and embeddings. **HIPAA-compliant RBAC**, enterprise-grade access controls, regulated data handling. **~60% retrieval time reduction.** Demonstrates building AI for a regulated enterprise domain with strict compliance requirements - the exact pattern for healthcare FDE deployments. *github.com/venkatalakshmiparimala/AI_on_FHIR*

CrashClarity: Predictive ML Pipeline - 85%+ Accuracy on 100K+ Records

Python · R · scikit-learn · Random Forest · Feature Engineering · GitHub Pages

Full ML lifecycle: feature engineering → model selection → evaluation → deployment. **85%+ accuracy.**

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, ML systems, experimental design, production data pipelines

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Bash, HTML/CSS

AI & Agents: LangChain, LangGraph, Multi-Agent Orchestration, ReAct / Plan-and-Execute, LLM APIs (Anthropic/Claude, OpenAI, Groq, Ollama), Prompt Engineering, Context Engineering, RAG Pipelines, ChromaDB, Vector Search, LLM-as-Judge Evaluation, CI/CD Eval Gates

Backend & Infra: FastAPI, Flask, React, PostgreSQL, MySQL, Docker, AWS (S3, EC2, RDS, Glue), Azure, CI/CD (GitHub Actions), RabbitMQ, Microservices

ML & Data: PyTorch, scikit-learn, BERT, spaCy, Hugging Face, NumPy, Pandas, Feature Engineering, Statistical Modeling

NLP & Healthcare: Clinical NLP (BERT, spaCy), FHIR API integration, HIPAA-compliant RBAC, PHI data handling

Process: Open-Source Development, TDD/BDD, Integration Testing, Git, JIRA, Agile/Scrum, Claude Code, GitHub Copilot